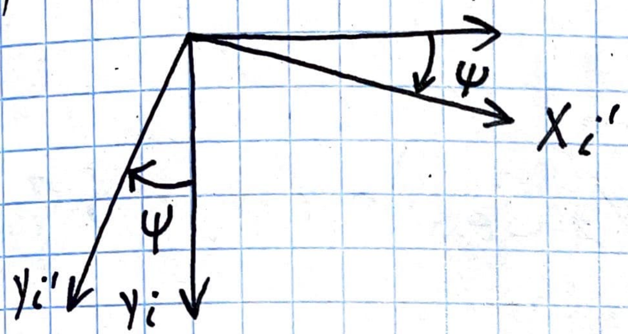
Связь углов Эйлера и кватернионов

• Для NED:

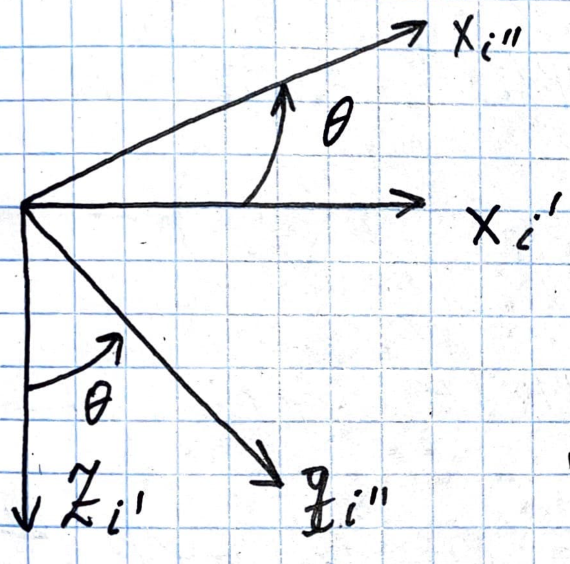
I)



$$R_i^{i'} = \begin{pmatrix} c_\psi & s_\psi & 0 \\ -s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Lambda^* = \cos\frac{\psi}{2} + i_3 \sin\frac{\psi}{2}.$$

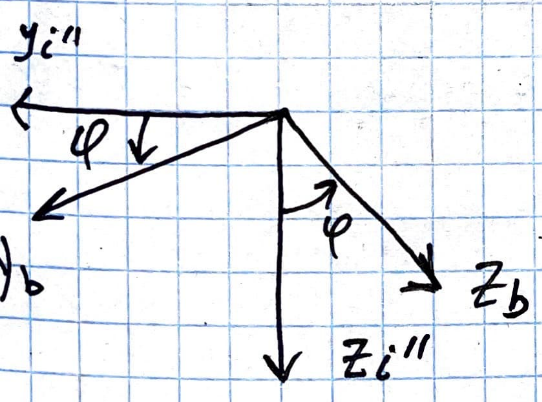
II)



$$R_{i'}^{i''} = \begin{pmatrix} c_\theta & 0 & -s_\theta \\ 0 & 1 & 0 \\ s_\theta & 0 & c_\theta \end{pmatrix}$$

$$M^* = c_{\theta/2} + i_2 s_{\theta/2}.$$

III)



$$R_{i''}^{b} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\varphi & s_\varphi \\ 0 & -s_\varphi & c_\varphi \end{pmatrix}$$

$$N^* = c_{\varphi/2} + i_1 s_{\varphi/2}.$$

Тогда согласно Th. о компонентах результирующего кватерниона в параметрах Родрига-Гамильтона:

$$Q^* = \Lambda^* \circ M^* \circ N^* =$$

$$= \left(C_{\psi/2} + i_3 S_{\psi/2}\right) \circ \left(C_{\theta/2} + i_2 S_{\theta/2}\right) \circ \left(C_{\varphi/2} + i_1 S_{\varphi/2}\right)$$

$$= \Big(C_{\psi/2} \cdot C_{\theta/2} + i_2 S_{\theta/2} \cdot C_{\psi/2} + i_3 C_{\theta/2} S_{\psi/2} -$$

$$- i_1 S_{\psi/2} \cdot S_{\theta/2} \Big) \circ \left(C_{\varphi/2} + i_1 S_{\varphi/2} \right) =$$

$$= C_{\psi/2} C_{\theta/2} C_{\varphi/2} + i_1 C_{\psi/2} C_{\theta/2} S_{\varphi/2} + i_2 S_{\theta/2} C_{\psi/2} C_{\frac{\varphi}{2}}$$

$$- i_3 S_{\theta/2} C_{\psi/2} S_{\varphi/2} + i_3 C_{\theta/2} S_{\psi/2} C_{\varphi/2} +$$

$$+ i_2 C_{\theta/2} S_{\psi/2} S_{\varphi/2} - i_1 S_{\psi/2} S_{\theta/2} C_{\varphi/2} + S_{\varphi/2} S_{\frac{\psi}{2}} S_{\frac{\theta}{2}}.$$

Компоненты кватерниона поворота из (NED) ~~одной~~ СК в другую, ~~вы~~ ~~ражени~~ Body Frame

$$q_0 = C_{\psi/2}\, C_{\theta/2}\, C_{\varphi/2} + S_{\psi/2}\, S_{\theta/2}\, S_{\varphi/2}.$$

$$q_1 = C_{\psi/2}\, C_{\theta/2}\, S_{\varphi/2} - S_{\psi/2}\, S_{\theta/2}\, C_{\varphi/2}$$

$$q_2 = C_{\psi/2}\, C_{\varphi/2} \cdot S_{\theta/2} + C_{\theta/2}\, S_{\psi/2}\, S_{\varphi/2} \qquad (2.15)$$

$$q_3 = C_{\theta/2}\, S_{\psi/2}\, C_{\varphi/2} - S_{\theta/2}\, C_{\psi/2}\, S_{\varphi/2}.$$

ψ - рыскание, θ - тангаж, φ - крен

От-но NED

$$\boxed{q_i^b = (q_0\ q_1\ q_2\ q_3)^T}$$